Lecture 4 - 13/05

Thursday, May 21, 2026 1:53 PM

linear systems - reminder

$\dot{\underline{x}} = A(t)\,\underline{x}$

1. fixexd point $x^* = 0$
   - can happen also if thers a $\lambda = 0$ eigenvalue, $AV = 0$, all the points on this eigenvector are FP.

2. for every initial condition $x(0)$, t can be driven forwards or backwards to get the trajectory, $-\infty < t < \infty$, this trajectory is unique

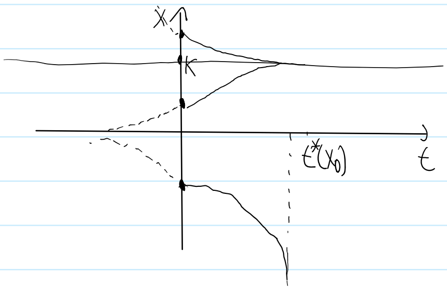
3. Same qualitative behavior everywhere. You can mult by contstant, but the trajectory will act the same

10 $\dot{x} = rx(- \frac{x}{K})$ $r, K > 0$

FP: $x = 0$, $x = K$

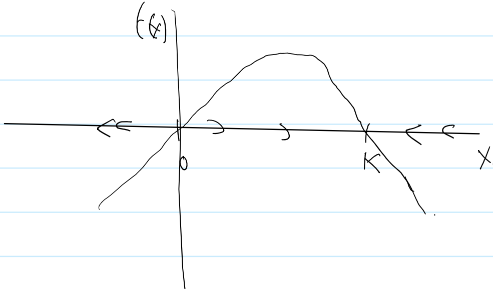
$\forall\, \underline{x}_0$, $x(t) = \frac{x_0 K}{x_0 + (K - x_0)e^{-rt}}$

looking at the different regions and how they act

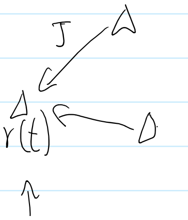


now we can look at $\dot{x}$:

Phase portrait:



neurons: cells that fire electrical signals, action potential
You can define the state of a neuron by defining the firing rate $r(t)$



a firing happens because other neuron activation so we can say a neurons ativation depends on the sum of the signa it recievs times some J_i, the strength of the connection, its input, call it h

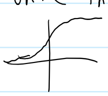
$$h(t) = \sum_{\substack{\text{neurons} \\ i}} r_i \cdot J_i$$

$\tau \dot{r} = -r + g(h)$

some time constant

$g(h)$ is usually some monotonic inc func

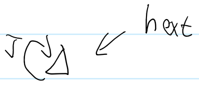
e.g. $g(h) = \frac{1}{1+e^{-\beta h}}$



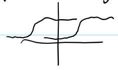
it we have 1 neuron that is connected to itself

$$\tau \dot{r} = -r + g(h) = -r + g(Jr) = -r + \frac{1}{1+e^{-\beta J r}}$$

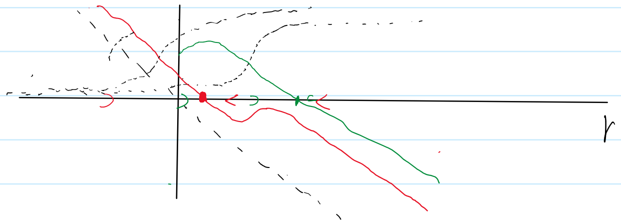
more generically, it can get some hext from others



$$\tau \dot{r} = -r + g(h) = -r + g(Jr + h_{ext}) = -r + \frac{1}{1+e^{-\beta(Jr+h_{ext})}}$$ ← hext here controls where the sigmoid curve happens

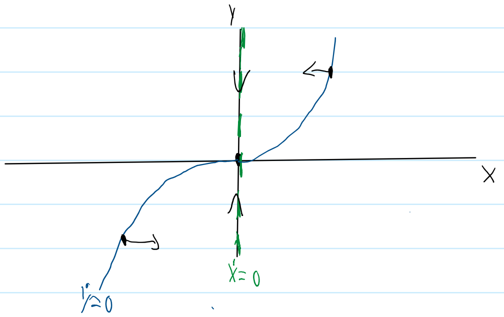


it has the linar $-r$ component
and a sigmoid one



without hext, stable point at 0, so it dies out
with hext, sigmoid is shifted an the stable point is positive

we should analyze the phase portrait to understand the interesting region



generalized in 2D

$$\begin{cases} \dot{x} = -x \\ \dot{y} = -2y + 2x^3 \end{cases}$$

first, find fixed points

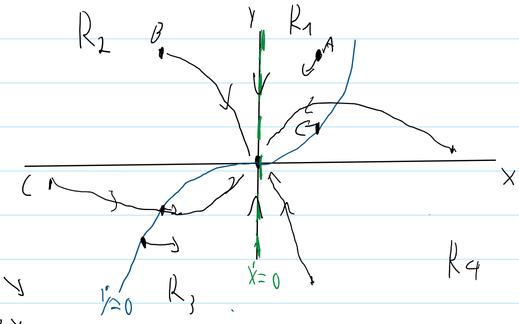
find lines where something changes about the flow, a generalization of a FP in 1D is a line in 2D, where one component is 0, called nullcline

any point on $x=0$ line, the flow will be vertical

look at $\dot{y}$ where $\dot{y}$ is 0, so $y = x^3$, any point on this line will go horizontal, only in x direction

now look at the direction of flow, a $x=0$, its $\dot{y} = -2y$, so it'll go towards (0,0)
on $\dot{y}=0$, $\dot{x} = -x$, so it'll go towards $x=0$

now look at the direction of flow, on $x=0$, its $\dot{y} = -2y$, so it'll go towards $(0,0)$
on $\dot{y}=0$, $\dot{x}=-x$, so it'll go towards $x=0$



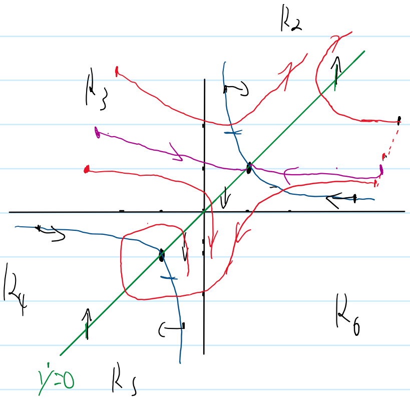
now we can divide the phase portrait into regions

look at A. by continuity, we can guess it'll go diagonal towards $(0,0)$.
intuition - why is this true! A couldn't go to the right ↘
if it did, there would be a point where it goes between ↙ & ↘
& must be ↓, but thats $\dot{x}=0$.

looking at R_2, flow should be ↘, since there is a line pointing away from the region on $\dot{y}=0$
some points (B) can go directly to $(0,0)$, and some (C) can cross to R_3.
R_3 is mirror of R_1, R_4 is mirror of R_2 (maybe B is fake and there only C, he wasnt sure)

another example

$$\begin{cases} \dot{x} = -x+y \\ \dot{y} = xy-1 \end{cases}$$



1) find FP, $(-1,-1)$, $(1,1)$

2) look for nullcline, $\dot{x}=0 \Rightarrow y=x$, points go vertical
$\dot{y}=0 \Rightarrow y=\frac{1}{x}$, points go horizontal

3) put arrows, for $\dot{x}=0$, $y=x$, so $\dot{y}=x^2-1$, so when $x^2>1$, $\dot{y}$ is +
for $\dot{y}=0$ $y=\frac{1}{x}$, so $\dot{x}=-x+\frac{1}{x}$, if x big, its $-x$, so towards 0
if x small, its $\frac{1}{x}$, so away
notice it isnt symmetrical, on the bottom theres some rotation thing, but not on top

4) look at regions, $R_1 \to$ either go up to R_2 or down to R_6, by continuity, theres some place that goes directly to FP$(1,1)$

$R_2 \to$ only goes up and right to (∞,∞)

$R_3 \to$ can cross to R_2, can cross to R_6 or go to FP $(1,1)$

$R_6 \to$ must go to R_5

then there is some rotation $\overset{\curvearrowleft}{R \to R \to R \to R}$

R_6 - must go to R_5

then there is some rotation $\overset{\curvearrowleft}{R_5 \to R_4 \to R_3 \to R_6}$
within the spiral, it cant spiral outwards (itll hit R_2 / R_1), so it either spirals in to FP(-1,1), or there is some elipse that is a stable orbit and out of it it spirals inwards, to it. within it it spirals outwards, to it

definitions

- Assymptotically stable FP - if all points around it, go to it
- Basin of attraction - the region from which trajectories go to FP
  - for a stable FP, BA = $\mathbb{R}^2$ (the full space)
- stable manifold of a FP $\underline{x}^*$ is defined:
  - $\{P_0 : \varphi(t;P_0) \xrightarrow[t\to\infty]{} \underline{x}^*\} = W_s(x^*)$

    where $\varphi(t,x_0)$ is trajectory with x_0 starting point
- unstable manifold - $W_u(\underline{x}^*) = \{P_0 : \varphi(t;P_0) \xrightarrow[t\to-\infty]{} \underline{x}^*\}$

FP (-1,1) is stable
FP (1,1) is unstable, if you are not direlty on a the stable manifold, you will go away

5) to decide what happens around FP, we can approximaty linearly

$\dot{\underline{x}} = \underline{F}(\underline{x})$ FP $\underline{x}^*$: $F(x^*) = 0$
near x^*, we' can look at $x(t) = \underline{x}^* + \underline{u}(t)$, u is very small

$\dot{u}(t) = F(\underline{x}^* + \underline{u}(t)) = \underset{=0}{F(\underline{x}^*)} + D\underline{F}_{(x^*)} \cdot \underline{u}$, where D is a matrix, $D_{ij} = \frac{\partial F_i}{\partial x_j}(x^*)$

$\underset{n\times 1}{\dot{u}(t)} = \underset{n\times n}{DF_{(x^*)}}\, \underset{n\times 1}{\underline{u}}$

if the sytem is nice and differantiable, and D is not degenerate, $\dot{\underline{u}}$ acts like $\dot{\underline{x}}$

ⓐ $x^* = y^* = 1$ for any (x,y), $DF_{(x,y)} = \begin{pmatrix} -1 & 1 \\ y & x \end{pmatrix}$

$\begin{cases} x = 1 + u \\ y = 1 + v \end{cases}$

for (1,1) $DF_{(1,1)} = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$

$\begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = \underset{A}{\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}} \begin{pmatrix} u \\ v \end{pmatrix}$

$\varphi(\lambda) = \begin{vmatrix} -1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = \lambda^2 - 2$, $\lambda_{1,2} = \pm\sqrt{2}$ - saddle FP

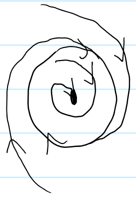
if τ = trace(A)

ⓑ $x^* = y^* = -1$ $\qquad DF_{(-1,-1)} = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$

$$\varphi(\lambda) = \begin{vmatrix} -1-\lambda & 1 \\ -1 & -1-\lambda \end{vmatrix} = \lambda^2 + 2\lambda + 2$$

since $\tau = -1$, $\Delta = 2$, FP is stable, so all points around it spiral inwards

(it is possible to still have an orbit



if $\tau = \text{trace}(A)$
$\Delta = \det(A)$
$\varphi(\lambda) = \lambda^2 - 2\tau + \Delta$
$\lambda = \dfrac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$

if $\tau < 0$ and $\Delta > 0$
the FP is stable since $\lambda > 0$